

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-001: Show / Hide Password Visibility Toggle in the Login Module

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle in the Login Module
Student Name	P.Tharanan
Student ID	JF/ICT/24/20
Implementation Status	Completed
Date	8 October 2026

1. Change Request Summary

1.1 Business Requirement

Users and administrators currently enter passwords into a masked input field without an option to view or verify entered text. As passwords often contain numbers and symbols, users frequently make typing errors without realizing it. The lack of visibility increases failed login attempts and repeated entries. The system should provide an intuitive password toggle allowing users to view or mask credentials before submitting.

1.2 Existing Behaviour

Before implementation:

- LoginPage.jsx renders LoginFormSection.jsx with username and password.
- The password input is statically configured with type="password".
- Password input characters are masked unconditionally with bullets.
- No visibility toggle button or eye icon exists to reveal entered text.
- Existing authentication sends credentials via POST /api/auth/login.

1.3 Requested Change

A client-side search facility was required above the administrator's student table.

The search must:

- Toggle password input type between "password" and "text".
- Default to masked/hidden state upon component mount and page load.
- Display intuitive eye and eye-slash visual indicator icons.
- Include accessible aria-label and aria-pressed attributes.
- Ensure toggle button uses type="button" to prevent form submission.
- Adjust layout in login.css so password text does not overlap the icon.
- Support keyboard activation via Space and Enter keys.
- Preserve existing login validation and error alert banners.
- Pass entered credentials accurately to login(username, password).
- Maintain existing administrator and student authentication flows.

2. Implementation

2.1 Implementation Summary

The CR-016 requirements were implemented across the authentication and student management modules to enhance user experience and security interface controls. A reactive visibility state was introduced to dynamically toggle the password field masking mechanism. The implementation integrates icon components from the lucide-react package (Eye and EyeOff) to provide intuitive, interactive visual cues within the password input fields.

2.2 Affected Files

File	Change	Purpose
frontend\src\components\LoginFormSection.jsx	Modified	Implemented a reactive password visibility toggle on the login page using Eye and EyeOff icons from lucide-react . This client-side enhancement allows administrators to seamlessly show or mask their input credentials without affecting existing validation or submission logic.

frontend\src\components\StudentFormModal.jsx	Modified	Implemented a reactive password visibility toggle on the admin dashboard add new student popup using Eye and EyeOff icons from lucide-react . This client-side enhancement allows administrators to seamlessly show or mask their input credentials without affecting existing validation or submission logic.
frontend\src\styles\login.css	Modified	Provides styling for the show/hide password eye icon and password field

Additional files: None / [List any additional files if modified]

2.3 Technical Changes

Password Visibility State

A React state variable was added to control the password visibility:
`const [showPassword, setShowPassword] = useState(false);`

The default value is false, ensuring that the password remains masked when the form is initially loaded.

Dynamic Password Input Type

The password input type is dynamically controlled:

```
type={showPassword ? 'text' : 'password'}
```

When showPassword is false, the password remains masked. When showPassword is true, the entered password becomes visible.

2.4 Implementation Logic

- Load the Login or Add Student form.
- Initialize showPassword as false.
- Display the password input using type="password".
- Display the eye icon. 5. User enters the password normally.
- User activates the eye button.
- Update showPassword to true.
- Change the input type to "text".

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend testing.

Testing Types:

- UI testing

Environment: Development

Browser: [Chrome] **Test Role:**

Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	Open Login page and inspect password field	Password is masked by default	Password is masked by default	Pass
TC02	Click Login eye icon	Password becomes visible	Password becomes visible	Pass
TC03	Click crose-eye icon	Password becomes hidden again	Password becomes hidden again	Pass
TC04	Open add new student popup and inspect password field	Password is masked by default	Password is masked by default	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	4
Passed	4
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Users can toggle password visibility between hidden and visible	UI testing	Met
AC-002	Default state of the password field on load is masked.	UI testing	Met
AC-003	Toggle icon updates dynamically to match visibility state.	UI testing	Met
AC-004	Activating the toggle does not trigger login form submission.	UI testing	Met
AC-005	Toggle button is keyboard accessible and includes aria-label.	UI testing	Met
AC-006	Password text remains legible and is not obscured by the toggle icon	UI testing	Met
AC-007	Valid credentials submitted in either state log in successfully.	UI testing	Met
AC-008	Existing validation errors and role-based redirects continue to work.	UI testing	Met

5. Evidence

A maximum of **two relevant screenshots** should be attached.

Evidence 01 – Login Password Visibility Toggle

Related: AC-001, AC-002

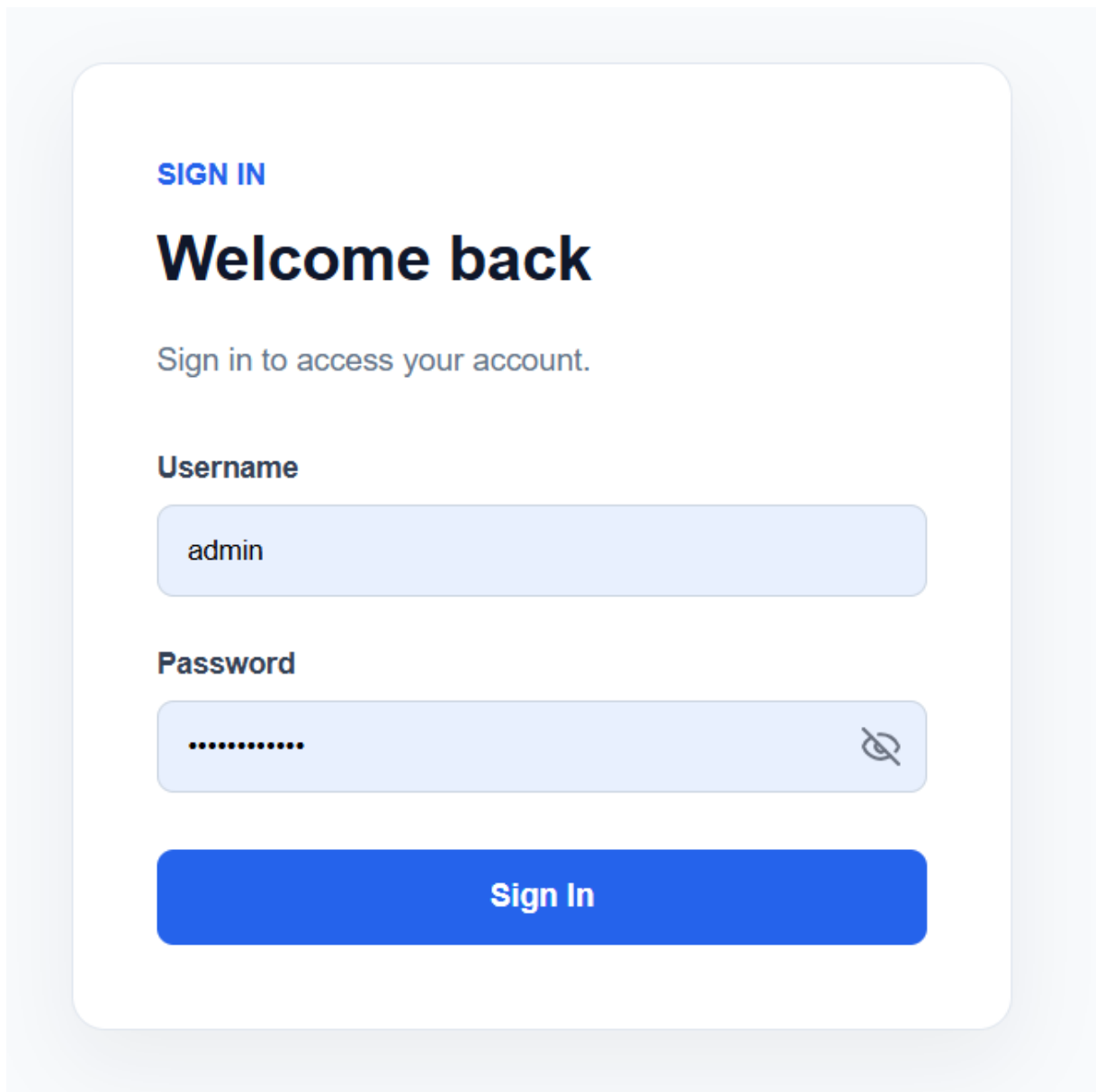
A screenshot of a login page. At the top, it says "SIGN IN" in blue. Below that is "Welcome back" in large, bold black text. Underneath is the instruction "Sign in to access your account." in a smaller, grey font. There are two input fields: "Username" with the text "admin" and "Password" with masked characters (dots). To the right of the password field is an eye icon, indicating the password is visible. At the bottom is a large blue button labeled "Sign In".

Figure 1: Login page showing the password field with the password visibility toggle. The screenshot should demonstrate the visible password state and eye icon.

Evidence 02 – Administrator Add Student Password Toggle

Related: AC-003, AC-005

The screenshot shows a web form titled "Add New Student" with a subtitle "Create a new student account." and a close button (X) in the top right corner. The form contains six input fields arranged in two columns: "Registration Number", "First Name", "Last Name", "Email", "Phone Number", and "Password". The "Email" field contains the text "admin". The "Password" field contains the text "Password123!". To the right of the password input is a visibility toggle icon, which is an eye with a slash through it, indicating that the password is currently hidden. At the bottom right of the form are two buttons: "Cancel" and "Create Student".

Figure 2: Administrator Add Student form showing the password visibility toggle and the visible password state.

6. Final Result

Item	Result
Requirements	Met / Partially Met / Not Met
Implementation	Completed / Partially Completed
Testing	Passed / Partially Passed / Failed
Acceptance Criteria	All Met / Partially Met / Not Met
Existing CRUD	Verified / Not Verified
Final Status	Accepted / Pending / Rejected

6.1 Final Summary

CR-016 was successfully implemented as a frontend password visibility enhancement for the Login and Administrator Add Student modules. The implementation allows users to show and hide entered passwords using an eye/close-eye toggle while keeping the password masked by default. Accessibility attributes, keyboard interaction, input tap, and form submission behaviour were addressed.

All defined acceptance criteria were verified and marked as met.

6.2 Issues / Limitations

[State any remaining issues or limitations.]

If none:

No significant issues identified.

Student Name: P.Tharanan

Student ID: JF/ICT/24/20

Date: 10 October 2026

Signature: P.Tharanan

~ END OF REPORT ~